

L CHAKKA

Backend Software Engineer — Python, AWS & Event-Driven Systems

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PROFESSIONAL SUMMARY

Backend Software Engineer with 3+ years' commercial experience building cloud-driven SaaS infrastructure, APIs and event-driven services in Python on AWS. Hands-on with relational databases (PostgreSQL, MySQL), DynamoDB, S3 and event-driven architectures (EventBridge, SQS/SNS), building third-party API integrations and production microservices for an enterprise client platform. Comfortable working in a small, fast-paced team, taking features from design through delivery, and keen to keep growing technical skills alongside experienced engineers.

KEY SKILLS

Databases: PostgreSQL, MySQL, DynamoDB (NoSQL), SQLite, SQL

Cloud & Storage (AWS): S3, Lambda, API Gateway, SQS, SNS, EventBridge, Step Functions, CloudWatch, KMS

Event-Driven & API Architecture: Event-driven architectures, third-party API integration, REST APIs, OpenAPI/Swagger, microservices

Programming Languages: Python, SQL, Shell Scripting

Infrastructure as Code & DevOps: CloudFormation, AWS SAM, Git, Bitbucket, GitHub Actions, Jenkins, CI/CD pipelines, Docker (familiar)

Testing & Quality: pytest, behave (BDD), unit test, Postman, code reviews, unit/integration/end-to-end testing

Data Structures & Algorithms: Core DS&A foundation from MSc coursework; ongoing practice via LeetCode/HackerRank

Frameworks & Libraries: FastAPI, Flask, Pydantic, boto3, Pandas, NumPy, PySpark

Ways of Working: Agile/Scrum, cross-functional collaboration, on-call/operational support, technical documentation

AI-Assisted Development: GitHub Copilot (code completion, test scaffolding, debugging support)

PROFESSIONAL EXPERIENCE

Software Engineer — Python & AWS Backend — Cognizant (Client: Direct Line Group, UK) Feb 2026 – Present

- Design and build scalable, cloud-driven microservices on AWS for a UK insurance partner onboarding platform, using S3 and DynamoDB for storage and event-driven services (EventBridge, SQS/SNS) to automate workflows, owning 6+ production services end-to-end.
- Developed a multi-type file ingestion service handling parsing, validation, data quality checks, reconciliation and secure S3 storage, exposed through REST APIs with OpenAPI/Swagger documentation and integrated with third-party partner systems.
- Built a Lambda-based automation service using ETag-based concurrency control for safe distributed writes to S3, integrated into a Step Functions state machine to remove manual intervention from a previously manual process.
- Built an event-driven monitoring service using EventBridge and SQS/SNS to detect and alert on missing data, replacing a manual daily monitoring process.
- Contributed core components to a distributed orchestration workflow (AWS Step Functions) coordinating processes across 5+ systems with retry logic and fault tolerance; resolved cross-account access issues in collaboration with DevOps.
- Authored the project's first Low Level Design (LLD) documentation standard, later adopted team-wide, and contributed to operational runbooks for production issue diagnosis and resolution.
- Participate in code reviews and design discussions; implemented a full testing strategy (unit, integration, end-to-end BDD) achieving 100% coverage on key components, and support production systems through monitoring and troubleshooting via CloudWatch.

Python Developer & Technical Consultant — Systemonex IT Solution Inc Jan 2025 – Dec 2025 | Remote, UK

- Designed and delivered Python-based technical solutions applying ETL principles, data processing techniques and REST API design patterns.
- Built Python code examples and project templates covering backend development, data pipelines and Pandas-based data validation, demonstrating real-world engineering best practice.

- Supported end-to-end project delivery, guiding software development lifecycle, debugging and data processing workflows.

Junior Software Engineer — YVR Technologies

Oct 2023 – Oct 2024

- Built Python-based data pipelines in collaboration with business stakeholders, improving data availability and supporting a 20% increase in data-driven decision-making.
- Implemented data validation logic using Pandas, reducing data discrepancies by 25%, and delivered analysis to senior management to identify operational efficiency improvements.
- Built and maintained RESTful API integrations enabling reliable data exchange between internal systems and third-party services.

Assistant System Engineer — Tata Consultancy Services (TCS)

Jul 2021 – Jul 2022

- Developed Python automation scripts to streamline business workflows, contributing to a 20% increase in operational efficiency.
- Wrote and executed test suites in Python and PyTest to identify defects early and reduce system downtime; collaborated with cross-functional teams on requirements analysis.

INTERNSHIPS

Data Science Intern — Exposys Data Labs

Jun 2020 – Jul 2020

- Applied statistical and machine learning methods for customer segmentation, improving marketing campaign targeting and recommendation accuracy.
- Delivered compelling presentations with actionable insights, influencing a 15% improvement in conversion rates and earning praise from internal teams and clients
- Led data pre-processing initiatives, achieving a 30% enhancement in model performance and optimizing predictive accuracy through rigorous cross-validation.

Machine Learning Intern — Indian Servers

Apr 2020 – Jul 2020

- Applied Python, NumPy and Scikit-learn to optimise machine learning models, reducing false positives by 15% through improved feature extraction.
- Built computer vision models using PyTorch and OpenCV, including transfer learning with ResNet for image classification.

EDUCATION

MSc Computer Science — University of East London, London, UK

Sep 2022 – Sep 2024

BTech Information Technology — VR Siddhartha Engineering College, India

Jun 2017 – Jul 2021

References:

1. Cognizant :- meiyarsu.selvam@cognizant.com – Tech lead
2. Systemonex - venkatanagadurga69@gmail.com – Manager